

# Simulating predictive-biomarker interactions in aggregated N-of-1 trials: a refined data-generating process following Hendrickson et al.

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## Abstract

**Background.** The power of aggregated N-of-1 trials to validate a predictive biomarker is estimated by simulation. Hendrickson et al. (2020) introduced a framework for this, and were, to our knowledge, the first in the N-of-1 literature to encode the biomarker-by-treatment interaction as a correlation between the biomarker and the drug response within a joint multivariate normal distribution. We examine that encoding, correct its defects, and specify a refined data-generating process for trials of prazosin-like drugs.

**Methods.** We derive the moments and power of a paired-difference interaction statistic in closed form under the published process and refined versions of it. We verify them by Monte Carlo simulation, including runs of the published code. We compare covariance-based and mean-based binding of the biomarker, and stress-test analysis models against the features the refined process adds.

**Results.** As soon as any carryover is present, the published coupling rule gives every off-drug visit the full on-drug biomarker correlation. The interaction vanishes, and power falls to near the test size at every positive half-life. Coupling that decays with the stated carryover half-life restores a smooth decline, monotone in the half-life for the paired-difference statistic, and raised power in the published code from 0.13 to 0.64 at a half-life of 0.1 week. Conditional on the biomarker, covariance-based binding is mean moderation with a reduced residual variance, so mean-based binding that fades with the drug effect loses power almost identically. Under compound symmetry the joint

distribution exists only for correlations below 0.256, so the published effect sizes of 0.3 and 0.6 relied on silently repaired matrices. Separable AR(1) correlation raises the bound to 0.44, or 0.48 at a serial correlation of 0.7. A biomarker that is also prognostic makes analyses that share the biomarker effect with baseline anticonservative.

**Conclusions.** The covariance-based encoding is coherent but needs graded coupling, a correlation structure with room for realistic effects, and validity checks that stop instead of repairing. We give the closed-form results that identify the power loss, the code changes for the published software, and a refined process for simulation studies of aggregated N-of-1 designs.

**Keywords:** N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

## Background

### Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times, and the individual series are analyzed together<sup>1-3</sup>. Because every participant contributes both an on-drug and an off-drug response, the design can estimate how the drug effect varies between people, and so test whether a baseline measurement predicts who benefits<sup>4,5</sup>. The motivating example is prazosin for post-traumatic stress disorder (PTSD). Higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo<sup>6</sup>, which makes blood pressure a candidate predictive biomarker<sup>7-9</sup>.

Closed-form power calculations exist only for simplified versions of these designs, so power is estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial<sup>10</sup>. The DGP decides what a simulation can show. In particular, it decides how the biomarker interaction is represented, how the drug effect builds and wanes, and how carryover from one period reaches the next.

### The framework of Hendrickson et al.

Hendrickson et al.<sup>11</sup> (hereafter RH) introduced the simulation framework this paper builds on. Its contributions include:

- **A component decomposition of the symptom score.** The score is the baseline level minus three responses, each following a modified Gompertz curve:
  - a biological drug response (BR), driven by cumulative time on drug;
  - an expectancy response (PB), driven by time in the trial and scaled by the participant’s expectancy of receiving drug;
  - a time-varying natural course (TV).
- **Design machinery.** It covers open-label, crossover, blinded discontinuation and a Hybrid design combining all three. It tracks time on drug, time since discontinuation and expectancy for each visit.
- **Open software** (the `pmsimstats` R package).

RH also encoded the predictive biomarker in a distinctive way. Simulation studies of N-of-1 and crossover designs usually represent treatment-effect heterogeneity in one of two ways. The first is random participant-by-treatment variation<sup>4,12–14</sup>. The second, for an observed moderator, is a biomarker-by-treatment term in the mean<sup>15–17</sup>. RH instead placed the biomarker in the same multivariate normal distribution as the response components. They gave it a nonzero correlation  $c_{bm}$  with the drug response at on-drug visits and zero correlation elsewhere. A predictive biomarker is then one whose association with the outcome differs between on-drug and off-drug visits. To our knowledge this was the first time the biomarker-by-treatment interaction was parameterized as a covariance in the N-of-1 literature. The parameterization keeps the marginal distribution of the drug response fixed and expresses the strength of the interaction as a correlation.

The encoding is natural for a biomarker that tracks individual variation in response rather than a fixed dose-response relation, and it lets a single correlation calibrate the strength of the effect against the variability of the drug response.

## Why a follow-up is needed

Reusing RH’s process for a series of simulation studies exposed two problems. Both trace to the implementation, not to the idea.

1. **The joint distribution often does not exist.** Under the compound-symmetry structure RH used, the correlation matrix is positive definite only for  $c_{bm}$  below a ceiling, 0.256 in the

Hybrid design without carryover. RH’s nonzero effect sizes were 0.3 and 0.6. The code repairs any non-positive-definite matrix silently. At those effect sizes 37% of the published grid’s matrices (20 of 54) were replaced by nearby valid matrices, whose correlations differ from the ones stated. At  $c_{bm} = 0.6$  the repair lowered the on-drug correlation to 0.54 to 0.57.

**2. Power collapses under any carryover.** RH reported that power to detect the interaction fell sharply once carryover was introduced. In the Hybrid design at  $N = 70$  and  $c_{bm} = 0.3$  it fell from 0.74 to 0.12 at a half-life of 0.1 week, when only 0.1% of the drug effect remains a week after stopping. We show that this is an artifact of the coupling rule, not a property of the designs.

This paper therefore has three aims:

- to identify, in closed form, why power collapses under the published process;
- to give the changes to the published code that remove both problems;
- to specify a refined DGP, consistent with the pharmacology of short-half-life, titrated drugs such as prazosin, intuitive in its parameters and flexible in the features it can represent, for use across simulation studies of aggregated N-of-1 designs.

## Methods

### The published data-generating process

We use RH’s notation, with a few additions. Participant  $i$  is observed at a baseline visit  $t = 0$  and at post-baseline visits  $t = 1, \dots, T$  (weeks 4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \text{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x; m, d, r) = m \frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

evaluated at a component-specific clock:

- **BR** at cumulative time on drug  $t_{od}$ , with  $m = 10.99$ ,  $d = 5$ ,  $r = 0.42$  and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks.
- **PB** at time in the trial, scaled by the expectancy  $e_t$  (1 in the open-label phase, 0.5 when blinded), with  $m = 6.51$ ,  $d = 5$ ,  $r = 0.35$  and SD  $10e_t$ .
- **TV** at time in the trial, with the same curve as PB and SD 10.

Off drug, the drug-response mean decays from its previous value with half-life  $t_{1/2}$ :  $\mu_{BR,t} = \mu_{BR,t-1} 2^{-t_{sd,t}/t_{1/2}}$ , where  $t_{sd}$  is the time since discontinuation.

The vector (biomarker, baseline,  $TV_{i.}$ ,  $PB_{i.}$ ,  $BR_{i.}$ ) is multivariate normal. Within each component the correlation between any two visits is  $\rho = 0.8$  (compound symmetry). Between components it is 0.2 at the same visit and 0.1 otherwise. The biomarker's correlation with  $BR_{it}$  is  $c_{bm} g_t$ , where  $g_t$  is the *coupling*. RH's rule sets  $g_t = 1$  wherever the drug-response mean is nonzero and 0 elsewhere.

RH analyzed each trial with the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept  $u_i$ , the binary drug indicator  $D_{it}$ , and baseline included as a row with  $D_{i0} = 0$ . The interaction  $\beta_{bm:D}$  is tested with Satterthwaite degrees of freedom.

## Estimand

The target is the on-drug biomarker moderation of the drug response: the change in the drug effect per unit of biomarker while the drug is fully present. Without carryover it equals  $\beta_{bm:D} = -c_{bm} \sigma_{BR} / \sigma_{bm}$  under covariance binding, and  $-\beta_{bm} \sigma_{BR} / \sigma_{bm}$  under mean binding.

Analyses that contrast on-drug with off-drug visits through a binary drug indicator estimate this quantity only when no moderation remains off drug. Under carryover they estimate an attenuated on-minus-off contrast. Part of the power loss under graded coupling therefore reflects the binary indicator in the analysis, not only the data-generating process. Analyses with an exposure-decayed indicator  $D_{bc,it}$  at the true half-life recover the on-drug moderation. The closed-form results and RH's analysis use the binary indicator; the stress test uses  $D_{bc,it}$ .

## Closed-form analysis

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores, which is regressed on  $B_i$  by least squares. We call it E9 when it uses post-baseline visits only, and E9b when baseline is counted as one more off-drug visit, as in RH's analysis. Monte Carlo simulation of RH's mixed model shows that E9b reproduces its mean estimates to within 0.003 and its power closely (Results). The derivation gives three results.

- **Proposition 1.** The variance of  $\Delta$  does not depend on the carryover half-life. The half-life enters only through the response means and the coupling, and neither appears in the variance.
- **Proposition 2.** The population slope is

$$\beta_{bm:D}^{\text{E9}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} (1 - \bar{g}_{\text{off}}), \quad \beta_{bm:D}^{\text{E9b}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \left( 1 - \frac{n_{\text{off}}}{n_{\text{off}} + 1} \bar{g}_{\text{off}} \right),$$

where  $\bar{g}_{\text{off}}$  is the mean coupling over the  $n_{\text{off}}$  post-baseline off-drug visits.

- **Power** is a noncentral- $t$  probability governed by the correlation between  $\Delta$  and  $B$ . That correlation is proportional to the *coupling gap*  $1 - \bar{g}_{\text{off}}$ .

So carryover can affect power only through the coupling gap. The propositions reduce the question of why power collapses to the question of what the coupling rule does to  $\bar{g}_{\text{off}}$ .

**The ceiling.** Write the response block of the correlation matrix as  $M$  and the biomarker's correlations with the response components as  $c_{bm}u$ . The full matrix is positive definite if and only if  $M$  is, and

$$c_{bm}^2 u^\top M^{-1} u < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory, so the ceiling on  $c_{bm}$  is  $(u^\top M^{-1} u)^{-1/2}$  (Appendix A). It depends on the correlation structure, the

design and the coupling profile.

## Refinements

Each refinement removes one defect (Table 1). The first three are corrections. The rest are modeling choices, and each can be switched off to recover the published process.

Table 1: Refinements to the published process.

| Step | Change                                                                                                     | Defect removed or feature added                                            |
|------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1    | Stop when the correlation matrix is not positive definite                                                  | Silent repair changes the correlations, and so the effect, without warning |
| 2    | Graded coupling: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, 0 before any exposure | Step coupling removes the interaction at any positive half-life            |
| 3    | Anchored carryover: decay the drug-response mean from its last on-drug value                               | The recursion compounds the decay at the second and later off-drug visits  |
| 4    | AR(1) correlation within each component, $\rho^{ t-s }$ in weeks                                           | Compound symmetry overstates long-lag correlation                          |
| 5    | Separable cross-component structure, $c_1 \rho^{ t-s }$                                                    | Raises the ceiling; gives a valid factored generator                       |
| 6    | Options: prognostic biomarker couplings, phase-specific measurement error, decay form, dropout             | Not representable in the published process                                 |

## Pharmacological grounding

The refined process is specified for short-half-life, titrated drugs, with prazosin as the reference case.

- **Time scales.** Prazosin's plasma half-life is a few hours, against weekly visits. The effect of a dose therefore wanes within days, and pharmacological carryover at the visit scale is short. We take half-lives of 0.1 to 0.2 week as the pharmacological range and regard longer carryover as behavioral or psychological, to be simulated as a labeled sensitivity scenario.

- **Titration and onset.** The drug response follows the Gompertz curve in time on drug, which reflects dose titration and the accumulation of benefit. Its landmarks (half of maximum at 4.7 weeks) should be read clinically when the parameters are chosen.
- **Re-exposure.** RH's process restarts time on drug at zero when the drug is resumed. For prazosin this is justified, because the drug is re-titrated after an interruption. For drugs without titration, the response should instead rebuild from its residual level. This is an option, not a default.
- **Coupling.** Graded coupling makes the biomarker's association with the drug response fade at the rate the drug effect fades. That is the pharmacological reading of a predictive biomarker: it predicts the size of the drug effect, so it cannot predict a drug effect that is no longer present.

## Covariance-based and mean-based binding

The biomarker can be bound to the drug effect in the covariance, as RH did, or in the mean. Under mean moderation, the drug response is shifted by  $\beta_{bm} b_i \sigma_{BR} h_t$ , where  $b_i$  is the standardized biomarker and  $h_t$  the exposure profile. Plain mean moderation uses  $h_t = 1$  on drug and 0 off drug; decayed mean moderation uses  $h_t = 2^{-t_{sd,t}/t_{1/2}}$  off drug, the profile of graded coupling. We compare the forms in the Discussion.

## Reference specification

Table 2 collects the refined process. Values not listed are RH's.

Table 2: The refined data-generating process. Defaults are those used in the stress test.

| Element                      | Specification                                                             | Default         |
|------------------------------|---------------------------------------------------------------------------|-----------------|
| Components                   | $Y_{it} = BL_i - (TV_{it} + PB_{it} + BR_{it})$ , Gompertz means as in RH | RH's parameters |
| Within-component correlation | AR(1) in weeks, $\rho^{ t-s }$                                            | $\rho = 0.7$    |

| Element                       | Specification                                                                             | Default                                         |
|-------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------|
| Cross-component correlation   | separable, $c_1 \rho^{ t-s }$                                                             | $c_1 = 0.2$                                     |
| Biomarker coupling            | $c_{bm}$ on drug; $c_{bm} 2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure | $c_{bm} \leq \text{ceiling}$                    |
| Carryover on the mean         | decay from the last on-drug value, $\mu_{BR, \text{last}} 2^{-t_{sd}/t_{1/2}}$            | $t_{1/2} = 0.1$ to 0.2 week (pharmacological)   |
| Re-exposure                   | time on drug restarts at zero (re-titration)                                              | on, for titrated drugs                          |
| Validity                      | stop if the correlation matrix is not positive definite                                   | always                                          |
| Prognostic couplings (option) | correlation of the biomarker with TV or PB; or mean moderation of TV, $\beta_{bm}^{TV}$   | off; 0.3 in the stress test                     |
| Measurement error (option)    | independent, SD by phase                                                                  | off; 3 open label, 6 blinded in the stress test |
| Decay form (option)           | exponential or Weibull with the same half-life                                            | exponential                                     |
| Dropout (option)              | monotone, completely at random or at random given the last score                          | off; 20% in the stress test                     |

## Simulation studies

Three sets of simulations verify the results. All are reproducible from the scripts listed in the Availability section.

- Closed-form verification.** 5,000 Monte Carlo replicates per cell of the paired-difference statistic, and 500 to 1,000 replicates of RH's mixed model, in the Hybrid, crossover and open-label plus blinded discontinuation designs at  $N = 70$ .
- The published code itself.** RH's `generateData()` at commit 58b32a9 was run twice on its own pipeline: as published, and with only the coupling lines replaced by graded coupling (Appendix B). We used the Hybrid design,  $N = 70$ ,  $c_{bm} = 0.25$ , no censoring, 500 replicates per cell, and the same seeds for both versions.

3. **A pre-registered stress test of analysis models** against the refined process. It has 39 cells and 26,500 simulated trials. Correlation structure, sample size, carryover, decay form, dropout, measurement error and a prognostic biomarker were each varied from a base cell (separable AR(1) at  $\rho = 0.7$ ,  $N = 70$ ,  $t_{1/2} = 0.5$  week). Size was judged at 1,000 null replicates per cell.

## Results

### The published coupling rule removes the interaction

Under RH's rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so its coupling is 1. Hence  $\bar{g}_{\text{off}} = 1$  and, by Proposition 2,

$$\beta_{bm:D}^{\text{E9}} = 0, \quad \beta_{bm:D}^{\text{E9b}} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \frac{1}{n_{\text{off}} + 1} \quad \text{for every } t_{1/2} > 0.$$

Power is a step function of the half-life (Table 3, Figure 1). It takes its no-carryover value at  $t_{1/2} = 0$  and a constant at any positive half-life, whether 0.01 week or 10. For E9b that constant is 0.081, a residue of the baseline row, which alone keeps zero coupling. The published analysis gives the same: the mean estimate falls from  $-0.126$  to between  $-0.026$  and  $-0.032$  at every positive half-life, and power from 0.63 to 0.08 to 0.11.

With constant coupling, the biomarker moderates the on-drug and off-drug responses equally, so there is no interaction left to detect. RH's finding that small carryover destroys power is therefore a property of the coupling rule, not of carryover or of the designs.

### Graded coupling restores a smooth decline

Under graded coupling the mean off-drug coupling is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t_{\text{off}}} 2^{-t_{sd,t}/t_{1/2}},$$

Table 3: Closed-form power by carryover half-life (weeks). Hybrid design,  $N = 70$ ,  $c_{bm} = \beta_{bm} = 0.25$ , compound symmetry. Monte Carlo estimates agree within simulation error.

| Coupling rule    | Statistic | 0     | 0.1   | 0.2   | 0.5   | 1     | 2     |
|------------------|-----------|-------|-------|-------|-------|-------|-------|
| Step (published) | E9        | 0.721 | 0.050 | 0.050 | 0.050 | 0.050 | 0.050 |
|                  | E9b       | 0.654 | 0.081 | 0.081 | 0.081 | 0.081 | 0.081 |
| Graded           | E9        | 0.721 | 0.721 | 0.712 | 0.631 | 0.473 | 0.278 |
|                  | E9b       | 0.654 | 0.653 | 0.647 | 0.586 | 0.473 | 0.328 |
| Mean moderation  | E9        | 0.681 | 0.681 | 0.682 | 0.682 | 0.682 | 0.682 |
|                  | E9b       | 0.619 | 0.619 | 0.620 | 0.620 | 0.621 | 0.621 |

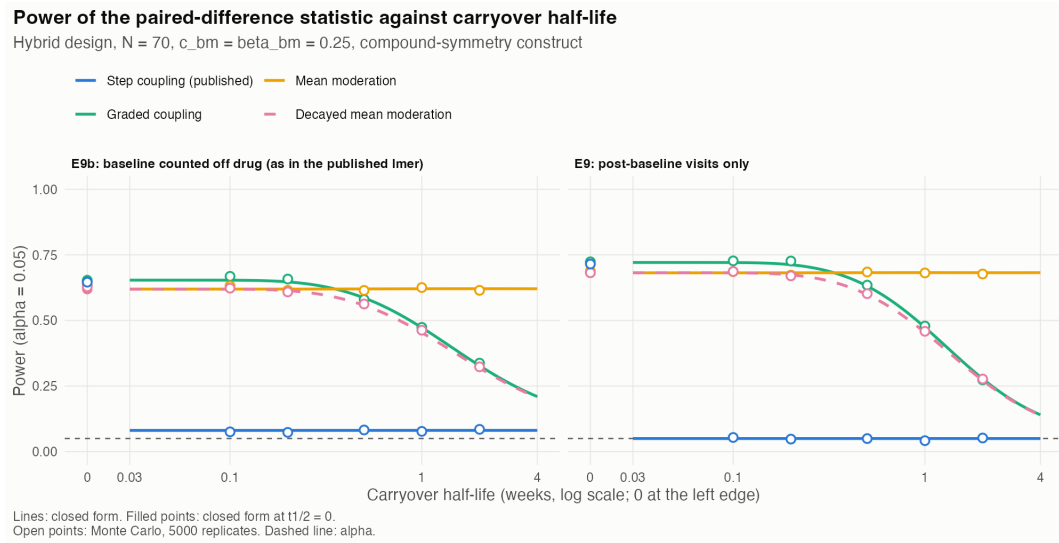


Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points), under the published step rule, graded coupling, mean moderation and decayed mean moderation. Hybrid design,  $N = 70$ , compound symmetry.

which is strictly increasing in  $t_{1/2}$ , because every off-drug visit has  $t_{sd} \geq 1$  week. Two consequences follow.

- **Power is strictly decreasing in the half-life** for the path-stratified paired-difference statistic (Proposition 3, Appendix A). For the pooled statistic it is non-increasing at every step of a 120-point grid from 0.02 to 4 weeks. For RH's mixed model it was checked only at the simulated half-lives.
- **The loss vanishes faster than any power of the half-life as  $t_{1/2} \rightarrow 0$ .** At 0.1 week,  $\bar{g}_{\text{off}} \leq 2^{-10}$  and power is unchanged to within 0.001.

The published rule is the limit of graded coupling as  $t_{1/2} \rightarrow \infty$ , applied at every positive half-life.

Power depends on the coupling profile only through the coupling gap  $1 - \bar{g}_{\text{off}}$  (Figure 2). The gap is 1 without carryover; under graded coupling it is 0.991 at 0.2 week, 0.906 at 0.5 and 0.754 at 1. Under the published rule it is 0 at every positive half-life. Because the noncentrality is proportional to the gap, the sample size needed for a given power scales roughly as  $1/\text{gap}^2$ .

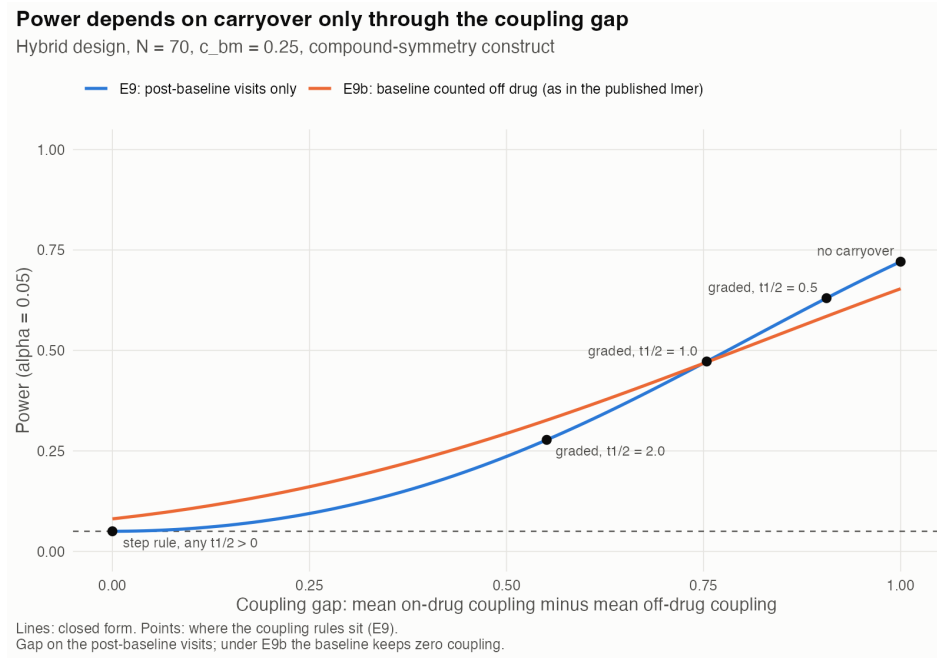


Figure 2: Closed-form power against the coupling gap, with the positions of the coupling rules. Hybrid design,  $N = 70$ ,  $c_{bm} = 0.25$ .

Table 4: RH’s code at commit 58b32a9, as published and with graded coupling: Hybrid design,  $N = 70$ ,  $c_{bm} = 0.25$ , 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022.

| $t_{1/2}$ (weeks) | Mean estimate, published | Mean estimate, graded | Power, published | Power, graded |
|-------------------|--------------------------|-----------------------|------------------|---------------|
| 0                 | -0.131                   | -0.131                | 0.668            | 0.668         |
| 0.1               | -0.028                   | -0.129                | 0.126            | 0.638         |
| 0.2               | -0.029                   | -0.131                | 0.092            | 0.656         |
| 1                 | -0.033                   | -0.107                | 0.094            | 0.462         |

## The correction in the published code

Replacing only RH’s coupling lines with graded coupling (Appendix B) and running their pipeline unchanged gave the following results (Table 4). The two versions coincide without carryover, as they must. At the published half-lives the graded version keeps the full interaction, and at 1 week it shows a gradual decline. No matrix was repaired in either version, because  $c_{bm} = 0.25$  is below the ceiling.

Figure ?? repeats RH’s Figure 4B on their own code, for all four designs, both sample sizes and the five censoring patterns. It uses  $c_{bm} = 0.25$  instead of RH’s 0.6, because 0.6 lies above every covariance-based ceiling and would confound the comparison with the repair. At 0.25 neither version repairs a matrix, so the coupling rule is the only difference.

[Results text for Figure 3 pending: the rerun of RH’s code was still in progress at this draft.]

## Room for larger effect sizes

The ceiling  $(u^\top M^{-1}u)^{-1/2}$  depends strongly on the correlation structure (Table 6).

- **Compound symmetry.** Under RH’s structure the ceiling is 0.256 in the Hybrid design, below both of RH’s nonzero effect sizes.
- **AR(1) within components,** with cross-component correlations that decay with the lag, raises the ceiling modestly, to 0.291 at  $\rho = 0.8$ . The effect depends on the design: in the open-label design, where every visit is on drug, AR(1) lowers the ceiling, because a single correlation repeated across visits that share little demands more of the biomarker.
- **The separable form** makes the cross-component correlation follow the same AR(1) decay  $(c_1\rho^{|t-s|})$ , so the response block is a Kronecker product. The ceiling rises to 0.440 at RH’s

Table 5: Correlation matrices that are not positive definite,  $\rho = 0.8$ . Published grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs;  $t_{1/2} \in \{0, 0.1, 0.2\}$  week. Extended grid: the last three designs;  $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$  week. The published-process counts reproduce those of RH’s own code exactly.

| Grid (number of matrices)                                  | Published (step coupling, compound symmetry) | Refined (graded coupling, separable AR(1)) | Mean-based binding |
|------------------------------------------------------------|----------------------------------------------|--------------------------------------------|--------------------|
| Published Figure 4 grid, $c_{bm} \in \{0, 0.3, 0.6\}$ (81) | 20 (25%)                                     | 21 (26%)                                   | 0 (0%)             |
| Extended grid, $c_{bm} = 0.25$ (40)                        | 0 (0%)                                       | 0 (0%)                                     | 0 (0%)             |
| Extended grid, $c_{bm} = 0.3$ (40)                         | 12 (30%)                                     | 0 (0%)                                     | 0 (0%)             |
| Extended grid, $c_{bm} = 0.6$ (40)                         | 12 (30%)                                     | 30 (75%)                                   | 0 (0%)             |

$\rho = 0.8$  and to 0.474 at  $\rho = 0.7$  (0.480 in the Hybrid design).

The higher ceiling has a cost. At the same  $c_{bm}$ , AR(1) data carry less information about the interaction than compound-symmetric data: the closed-form power of the paired-difference statistic in the Hybrid design at  $c_{bm} = 0.25$  falls from 0.721 to 0.335. A larger feasible correlation does not by itself mean more power, so effect sizes under the two structures should be compared on the scale of the implied slope, not of  $c_{bm}$ .

Table 5 counts the correlation matrices that fail the published code’s own positive-definiteness test (`corpcor::is.positive.definite()`), which would trigger its silent repair. There is one matrix per design path, effect size and half-life;  $N$  does not enter the matrix. Two points stand out.

- At  $c_{bm} = 0.3$  the refined process needs no repair, whereas the published process repairs 30% of its matrices.
- At  $c_{bm} = 0.6$  the refined process fails more often than the published one. The published process passes those cells at positive half-lives only because its step rule makes the coupling constant after first exposure. A constant coupling is cheap to accommodate, and it carries no interaction. The cells that stop needing repair under the published rule are the cells that have lost their signal.

Only mean-based binding represents every cell.

A biomarker correlation of 0.45 is feasible in every design under the separable form at  $\rho = 0.7$ , and not under the published structure. Effects as large as RH’s 0.6 exceed every covariance-based

Table 6: Largest feasible biomarker correlation  $c_{bm}$  under graded coupling, without carryover, minimum over the Hybrid, crossover and open-label plus blinded discontinuation designs. Under the separable form at  $\rho = 0.7$  the Hybrid value is 0.480.

| $\rho$ | Compound symmetry | AR(1), decaying cross-correlation | Separable AR(1) |
|--------|-------------------|-----------------------------------|-----------------|
| 0.8    | 0.256             | 0.291                             | 0.440           |
| 0.7    | 0.346             | 0.452                             | 0.474           |

ceiling in the Hybrid design. They can be represented only by binding the biomarker in the mean.

### Consequences for the analysis

The refined process can simulate features the published one cannot, and these change which analysis is safe. Two results from the stress test bear on the DGP directly.

**A prognostic biomarker.** Suppose blood pressure is associated with the natural course or with placebo response as well as with the drug effect. We simulated a correlation of 0.3 with the natural-course component (constant over visits, or growing along its curve) or with the placebo component. Analyses that keep baseline as an outcome row use one biomarker main effect  $\beta_B$  for baseline and for the off-drug visits. When the biomarker is prognostic, those two associations differ, and part of the difference is attributed to the interaction (Appendix A.4).

Every analysis that shares the biomarker effect with baseline failed in at least one prognostic cell:

| Analysis                                     | Natural course,<br>constant | Natural course,<br>growing | Placebo response |
|----------------------------------------------|-----------------------------|----------------------------|------------------|
| Published (random intercept,<br>model-based) | 0.107                       | 0.053                      | 0.159            |
| Random intercept + CAR(1), CR2               | 0.099                       | 0.049                      | 0.100            |
| The same, half-life chosen by AIC            | 0.115                       | 0.058                      | 0.112            |
| Unstructured covariance,<br>Kenward-Roger    | 0.416                       | 0.195                      | 0.222            |

| Analysis                                                      | Natural course,<br>constant | Natural course,<br>growing | Placebo response |
|---------------------------------------------------------------|-----------------------------|----------------------------|------------------|
| The same, half-life<br>chosen by AIC                          | 0.509                       | 0.245                      | 0.256            |
| Random intercept +<br>CAR(1) with<br>expectancy terms,<br>CR2 | 0.047                       | 0.046                      | 0.040            |
| Random intercept +<br>CAR(1) with free<br>baseline slope, CR2 | 0.049                       | 0.063                      | 0.049            |

*Null rejection rates, 1,000 replicates per cell.*

The two analyses that free the biomarker's baseline association held the nominal level in all three cells. One adds expectancy and a biomarker-by-expectancy term to the mean model; the other adds `bmc:b10`, which gives baseline its own biomarker slope.

**Size more generally.** Pooled over 11 non-prognostic null cells (11,000 replicates; nominal band 0.046 to 0.054):

- the published analysis rejected 0.069, and 0.081 at  $N = 35$ ;
- an unstructured model with Kenward-Roger inference<sup>18</sup> rejected 0.057, with 0.087 at  $N = 35$ ;
- random-intercept models with continuous-time AR(1) residuals and CR2 cluster-robust standard errors<sup>19,20</sup> rejected 0.048 to 0.053, with no single cell above 0.069.

The pre-registered selection rule chose the expectancy-term model as the only analysis that passed every screen. One screen, for power symmetry in the sign of the coupling, proved defective: it tested a property that holds exactly by construction, so its failures were Monte Carlo error. Dropping it does not change which analyses are both nominal and robust to a prognostic biomarker. Its power cost is about a third under AR(1) data and small under compound symmetry.

A simulation that omits a prognostic pathway cannot reveal these failures. The refined process includes it as an option for that reason.

## Discussion

### Covariance-based and mean-based binding

**What the two encodings share.** Under joint normality, conditioning on the biomarker shows how the covariance encoding relates to mean moderation. With  $b_i$  the standardized biomarker,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm} g_t \sigma_{BR} b_i, \quad \text{Cov}(BR_{it} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2 g g^\top.$$

The covariance encoding is therefore mean moderation with exposure profile equal to the coupling profile  $g_t$  and strength  $\beta_{bm} = c_{bm}$ . It differs only in that its residual drug-response covariance is reduced by a rank-one term. Three consequences follow.

- **The power loss under carryover is set by the profile.** Mean moderation with the same fading profile loses power almost identically (below), and this is a consequence of the identity, not an independent finding.
- **The small power advantage of the covariance encoding** (0.721 against 0.681 for E9 without carryover) comes from the rank-one reduction of the residual variance. It is a matter of how the variance is parameterized, not of the encoding.
- **The ceiling is the condition** that the reduced residual covariance remain positive definite.

RH's contribution is therefore a parameterization of the interaction, as a correlation that keeps the marginal variance of the drug response fixed, rather than a different model of the mean. That parameterization is what was new in the N-of-1 literature, to our knowledge, and both its strengths and its defects follow from it.

RH's covariance-based encoding has real strengths.

- **One interpretable parameter.** The interaction strength is a single correlation, interpretable as the share of drug-response variation the biomarker explains.
- **A reduced form of responder heterogeneity.** If participants fall into latent responder classes and the biomarker is associated with class membership, the second moments of the resulting mixture are those of the covariance encoding, with  $c_{bm}^2$  equal to the product of the between-class variance fractions of the biomarker and the drug response<sup>21</sup>. The encoding

therefore approximates a responder-class mechanism without requiring the mixture to be identified. Identifying it needs samples far larger than an N-of-1 program has. The approximation has limits: no two-class mixture reaches the covariance encoding exactly, because a mixture always moves the conditional mean as well.

- **The marginal variance of the drug response is fixed.** The biomarker explains part of the existing variation rather than adding to it, so the effect size can be stated as a share of response variance without changing the response distribution.
- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

Its weaknesses, all made explicit by this analysis, are structural.

- **The joint distribution must exist**, which bounds the effect size below values that are clinically plausible. The bound depends on the correlation structure and the design.
- **The coupling must be specified at off-drug visits**, and the natural choice of an indicator is wrong under carryover.
- **The drug-response component keeps its full variance at off-drug visits**, with zero mean. This adds noise to every on-off contrast, and it is the least defensible feature of the published construct.
- **A correlation is less intuitive than a mean effect for clinical readers.** The slope it implies,  $-c_{bm}\sigma_{BR}/\sigma_{bm}$ , must be derived before it can be compared with an observed effect. Even then the comparison needs a common outcome scale. The reported association for prazosin, 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg of standing systolic pressure<sup>6</sup>, is on a scale the simulated symptom score does not share. Calibrating RH's parameters to such data is future work.

Mean moderation has no ceiling, an intuitive parameter (points of symptom change per SD of biomarker) and no coupling to specify off drug. In its plain form the shift is applied only at on-drug visits, so its power is flat in the half-life (0.681 at every value for E9). But it adds variance instead of explaining it, and that flatness is a coding choice, not a finding: the biomarker's effect stops the moment the drug is stopped, while the drug effect itself carries over.

When the mean shift is decayed with the drug effect, as graded coupling decays the correlation, power declines almost exactly as it does under graded coupling: 0.681, 0.681, 0.674, 0.598, 0.454

and 0.271 at half-lives of 0, 0.1, 0.2, 0.5, 1 and 2 weeks, against 0.721, 0.721, 0.712, 0.631, 0.473 and 0.278. As the identity above implies, how much carryover costs is decided by the exposure profile of the interaction, whether it fades with the drug effect, and not by whether it is bound in the covariance or in the mean. Comparisons that report larger carryover losses under covariance binding than under mean binding are comparing a fading interaction with a non-fading one.

A natural next step combines the strengths. Generate the drug response from a participant-level sensitivity to exposure,

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \quad \omega_i = \bar{\omega} (1 + \zeta b_i) + u_i,$$

where  $f(t)$  is the exposure-response time course,  $\zeta$  the biomarker's effect on sensitivity and  $u_i$  the sensitivity it does not explain. This is a random-coefficient model with a covariate-by-exposure interaction, the form used to describe individual response in replicated crossover and N-of-1 trials<sup>4,12</sup>.

- It binds the biomarker in the mean through individual sensitivity, so it has no ceiling.
- Its moderation is proportional to the response and fades with it.
- Its off-drug variance shrinks with the response.

It is not yet implemented in the software, and its response half-life for prazosin in PTSD is not known.

## Recommendations

For simulation studies that build on RH, we recommend the following.

1. **Always apply the corrections:** stop on non-positive-definite matrices, use graded coupling, and use anchored carryover.
2. **Use the separable AR(1) structure** at a stated  $\rho$ , and choose  $c_{bm}$  below the ceiling for every design in the study.
3. **Keep RH's process as a reference arm** whenever results are compared with the published ones. Attribute each difference to a named refinement.
4. **Report effect sizes on two scales:** as  $c_{bm}$ , and as the implied slope in outcome points per SD of biomarker.

5. **Include a prognostic scenario** whenever the analysis shares the biomarker effect with the baseline visit.
6. **Use mean moderation, or the sensitivity model once available, for effects above the covariance ceiling**, and as a sensitivity analysis on the encoding itself.

## Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for RH's mixed model. The two agree closely in the Hybrid design, but in the crossover design the closed form understates what a time-adjusted analysis achieves.
- **Parameter values are not estimated.** The Gompertz parameters, correlations and prognostic couplings are RH's values or chosen values, not estimates from prazosin trial data.
- **The process is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.

## Conclusions

RH introduced a covariance-based encoding of the predictive-biomarker interaction that remains a coherent choice for simulating aggregated N-of-1 trials. Its published implementation removes the interaction whenever there is carryover, and it silently alters correlations it cannot represent. Both problems have closed-form explanations and small fixes. With graded coupling, a separable correlation structure and validity checks that stop instead of repairing, the process behaves as the pharmacology of a short-half-life drug suggests it should: power declines smoothly with the half-life, and negligibly at pharmacological half-lives. We propose the refined process, with options for prognostic pathways and other features that change which analysis is valid, as the reference for simulation studies of these designs, pending calibration of its parameters to trial data.

## Abbreviations

BR: biological drug response; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; DGP: data-generating process; OL+BDC: open label followed by blinded

discontinuation; PB: expectancy (placebo-belief) response; PTSD: post-traumatic stress disorder;  
RH: Hendrickson et al. (2020); TV: time-varying natural course.

## Declarations

**Ethics approval and consent to participate.** Not applicable: the study uses simulated data only.

**Consent for publication.** Not applicable.

**Availability of data and materials.** All code and simulated data are in the pmsimstats-ng repository:

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-form`
- the check of the published code, in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patch`
- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` and `12-strawman-stress-decision.R`;
- the published code, at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

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## Appendix A. Closed-form derivations

### A.1 Moments of the paired-difference statistic

Let  $a_t = 1/n_{\text{on}}$  at on-drug visits and  $-1/n_{\text{off}}$  at off-drug visits, so that  $\Delta_i = \sum_t a_t Y_{it}$ . Because the weights sum to zero, the baseline level cancels and  $\Delta = -\sum_t a_t S_t$ , with  $S_t = TV_t + PB_t + BR_t$ . With  $m_1 = \sum a_t$ ,  $m_2 = \sum a_t^2$ ,  $m_e = \sum a_t e_t$ ,  $m_{ee} = \sum a_t^2 e_t^2$  and  $m_{2e} = \sum a_t^2 e_t$ , the variance under compound symmetry is

$$\begin{aligned} \text{Var}(\Delta) = & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho m_1^2 + (1 - \rho)m_2] + \sigma_{PB}^2[\rho m_e^2 + (1 - \rho)m_{ee}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_{\times}m_1^2 + (c_1 - c_{\times})m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_{\times}m_1m_e + (c_1 - c_{\times})m_{2e}], \end{aligned}$$

where  $c_1$  and  $c_{\times}$  are the same-visit and other-visit cross-component correlations.

- **Proposition 1.** No term depends on  $t_{1/2}$ , so  $\text{Var}(\Delta)$  is free of the half-life. The argument applies to any construct in which the half-life does not enter the response block.
- **Proposition 2.** Only  $BR_t$  is correlated with  $B$ , with covariance  $c_{bm}g_t\sigma_{bm}\sigma_{BR}$ , so  $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm})\sum_t a_t g_t$ . With  $g_t = 1$  on drug, this gives the expressions in the Methods.
- **Residual variance.** Under covariance binding,  $\tau^2 = \text{Var}(\Delta) - \beta_{bm:D}^2\sigma_{bm}^2$ . Under mean binding the shift adds variance and  $\tau^2 = \text{Var}(\Delta)$ .

### A.2 Power

Within a path of  $n$  participants, the least-squares slope is exactly normal given the biomarker values. Its  $t$ -statistic is exactly noncentral  $t$  with  $n - 2$  degrees of freedom and noncentrality  $\beta_{bm:D}\sqrt{\text{SS}_B}/\tau$ , where  $\text{SS}_B \sim \sigma_{bm}^2\chi_{n-1}^2$ .

At  $\text{SS}_B = \sigma_{bm}^2(n - 1)$  the noncentrality is  $\Lambda = \sqrt{n - 1}\rho_{\Delta B}/\sqrt{1 - \rho_{\Delta B}^2}$ , where  $\rho_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\text{SD}(\Delta)$  is proportional to the coupling gap.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance; in the Hybrid design it is under 0.7% of the total. Power is the noncentral- $t$  tail probability averaged over the  $\chi^2$  law of  $\text{SS}_B$ .

### A.3 Monotonicity under graded coupling

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}} t_{1/2}^2} \sum_{t_{\text{off}}} t_{sd,t} 2^{-t_{sd,t}/t_{1/2}} > 0.$$

- **Proposition 3.** By Proposition 2 the slope's magnitude is strictly decreasing in  $t_{1/2}$ . By A.1,  $\tau^2$  is strictly increasing, since  $\text{Var}(\Delta)$  is fixed. So the noncentrality and the power are strictly decreasing.
- **Small half-lives.**  $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$ , which gives the behavior near zero stated in the Results.
- **Large half-lives.**  $\bar{g}_{\text{off}} \rightarrow 1$  as  $t_{1/2} \rightarrow \infty$ , which recovers the step rule.

### A.4 The ceiling and the shared biomarker effect

**The ceiling.** Order the variables as biomarker, then responses, and partition the correlation matrix as  $\begin{pmatrix} 1 & c_{bm} u^\top \\ c_{bm} u & M \end{pmatrix}$ . By the Schur complement, the matrix is positive definite if and only if  $M$  is positive definite and  $1 - c_{bm}^2 u^\top M^{-1} u > 0$ . Hence the ceiling is  $c_{bm}^* = (u^\top M^{-1} u)^{-1/2}$ .

**The shared biomarker effect.** Let  $s_0$ ,  $s_{\text{off}}$  and  $s_{\text{on}}$  be the true slopes of  $Y$  on  $B$  at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one  $\beta_B$  for  $s_0$  and  $s_{\text{off}}$ .

The following account is heuristic, not a derivation. Consider a prognostic biomarker with no interaction, whose post-baseline slope is constant:  $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$ .

- **The leak.** The fitted  $\hat{\beta}_B$  lies between the baseline and off-drug slopes,  $\hat{\beta}_B \approx f s_{\text{prog}}$  for some  $0 < f < 1$  that depends on the weight the baseline row receives. The interaction absorbs the remainder,  $\hat{\beta}_{bm:D} \approx (1 - f) s_{\text{prog}} \neq 0$ .
- **The remedy.** Adding  $B_i \cdot \mathbf{1}\{t = 0\}$  frees  $s_0$  and removes this source of bias. That the baseline row imposes such a constancy constraint is known from the literature on baseline as a covariate or a response<sup>22</sup>.

Two further mechanisms operate in the stress-test scenarios.

- **The slope grows over time.** When the post-baseline slope grows, the bias depends on how drug status is confounded with time. Freeing  $s_0$  is then not sufficient in general. The residual 0.063 for that analysis is consistent with this.

- **Expectancy.** When the biomarker is associated with the placebo response, the slope also scales with the expectancy  $e_t$ , which is higher at the open-label (on-drug) visits. The expectancy term  $B_i \cdot e_t$ , with  $e_0 = 0$ , addresses both this mechanism and the baseline constraint.

A scenario in which the biomarker is correlated with baseline severity was not simulated.

## Appendix B. Code changes

### B.1 Graded coupling in RH's `generateData()` (commit 58b32a9)

The published lines (R/`generateData.R`, lines 131 to 138) couple the biomarker wherever the drug-response mean is nonzero:

```
# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p], "br", sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1, 'bm']<-modelparam$c.bm
    correlations['bm', n1]<-modelparam$c.bm
  }
}
```

The replacement couples in proportion to the remaining drug effect, with no coupling before first exposure:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (graded coupling), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p], "br", sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }
}
```

```

}else{
  g<-0
}
correlations[n1,'bm']<-modelparam$c.bm*g
correlations['bm',n1]<-modelparam$c.bm*g
}

```

## 506 B.2 Stop instead of repairing

507 The published code replaces a non-positive-definite matrix with a nearby valid one when its  
 508 `makePositiveDefinite` option is on, as it is in RH's driver (lines 145 to 150). The replacement  
 509 stops, so that an infeasible cell is reported rather than silently changed:

```

# published:
# if(makePositiveDefinite){
#   if(!is.positive.definite(sigma)){
#     sigma<-make.positive.definite(sigma, tol=1e-3)
#   }
# }

if(!corpcor::is.positive.definite(sigma)){
  stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
              modelparam$c.bm))
}

```

## 510 B.3 Anchored carryover on the drug-response mean

511 The published recursion (lines 83 to 92) decays the previous visit's mean by the total time since  
 512 discontinuation. At the second and later off-drug visits that mean has already been decayed, so the  
 513 decay compounds. The replacement decays from the last on-drug value:

```

brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
}

```

```

for(p in 1:nP){
  if(d[p]$onDrug){
    last_on<-brmeans[p]
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
  }
}
}

```

## B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every visit. In the pmsimstats-ng carryover manuscript this guard was missing. In a paired rerun of that manuscript's crossover cells (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`,  $N = 70$ ,  $c_{bm} = 0.45$ ,  $t_{1/2} = 1$  week, 500 replicates), restoring the guard raised the power of the exposure-weighted analysis from 0.512 to 0.814, the level of the binary indicator (0.810):

```

exposure_weight <- function(Db, tsd, halflife) {
  w <- 0.5^(tsd / halflife)
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))
}

```

## B.5 The refined process in pmsimstats-ng

The package function `buildSigma()` (`R/generateData.R`) implements graded coupling, with `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It also implements the AR(1) structures and the placebo coupling `c.bm.pb`. Two parts of the refined process are not yet in the package:

- **The repair is still there.** Its sampler still repairs a matrix silently when the Cholesky factorization fails (`R/generateData.R`, lines 390 to 392). That repair must become an error,

as in B.2.

- **The remaining refinements are in the drivers only.** The separable form, anchored carryover, and the natural-course and measurement-error options are implemented in the simulation drivers of the stress test and are to be moved into the package.